

CS164 Final Project: Sustainable Crop Rotation Optimization

1 Problem Description & Context

For this assignment I wanted to use linear programming to solve a problem specifically in sustainability. Sustainability has been a problem that draws me, and I try to incorporate it in my work at Minerva whenever I can (Also, I just got my dream internship at a sustainability company, so it is inspired by that!)

I found this recent study published in Nature Communications (Yang et al., 2024) that ran a 6 year field experiment in the North China Plain. This area produces like 23% of China's total cereal food. The traditional farming system used there is this winter wheat and summer maize double cropping rotation (WM). Now this is super productive for grain but it's also not so good on the environment mostly because of high greenhouse gas emissions and degraded soil health. I will be focusing on the GHG for this assignment.

The researchers tested 3 alternative crop rotations by introducing cash crops (sweet potato) and legumes (peanut, soybean). What was extra nice about this study is that they gave me all the numbers for the crop productivities in terms of crop yields, economic returns, GHG emissions, and cereal calorie production. This is what I will use for my problem formulation.

My optimization problem is as follows: A farmer named Bob has a limited amount of land and needs to figure out how to allocate across the different farming systems to maximize profit while maintaining enough cereal production (food security!) and staying under an emissions cap (climate goals!).

1.1 Data from the Field Study

The 6-year experiment (2016-2022) was at the Luancheng Station in Hebei Province. I normalized everything below with WM as the baseline = 1.00.

Rotation	Econ benefit	Cereal calories	Net GHG
WM (baseline)	1.00	1.00	1.00
SpWM (sweet potato)	1.60	0.69	0.17
PWM (peanut)	1.18	0.65	0.08
SWM (soybean)	1.18	0.65	0.10

2 Problem Formulation

2.1 Decision Variables

Let $x = [x_1, x_2, x_3, x_4]^T \in \mathbb{R}^4$ represent the fraction of land allocated to:

- x_1 : Winter Wheat–Summer Maize (WM)
- x_2 : Sweet Potato→Wheat–Maize (SpWM)
- x_3 : Peanut→Wheat–Maize (PWM)
- x_4 : Soybean→Wheat–Maize (SWM)

where $x_i \in [0, 1]$.

2.2 Objective Function

Our goal is to maximize total economic benefit for farmer Bob relative to baseline:

$$\max \quad Z = 1.00x_1 + 1.60x_2 + 1.18x_3 + 1.18x_4$$

2.3 Constraints

Land Allocation: Our first constraints is that all land must be allocated.

$$x_1 + x_2 + x_3 + x_4 = 1$$

Cereal Calorie Production: We impose a constraint that the government wants to maintain at least 80% of baseline cereal production for food security.

$$1.00x_1 + 0.69x_2 + 0.65x_3 + 0.65x_4 \geq 0.80$$

This makes our solution Non Trivial as we force some cereal-heavy systems in the mix.

GHG Emissions Cap: To meet sustainable goals we aim to achieve at least 50% reduction in emissions.

$$1.00x_1 + 0.17x_2 + 0.08x_3 + 0.10x_4 \leq 0.50$$

Non-negativity: Note we also do not allow for $x_i > 1$ however this constraint is already satisfied by the land and the non negativity constraint combined.

$$x_i \geq 0 \quad \forall i$$

2.4 Complete Linear Program

$$\begin{aligned} \max_x \quad & Z = 1.00x_1 + 1.60x_2 + 1.18x_3 + 1.18x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 = 1 \\ & 1.00x_1 + 0.69x_2 + 0.65x_3 + 0.65x_4 \geq 0.80 \\ & 1.00x_1 + 0.17x_2 + 0.08x_3 + 0.10x_4 \leq 0.50 \\ & x_i \geq 0 \quad \forall i \end{aligned}$$

3 Solution Approach: Three Different Norms

Instead of just solving the standard LP, I wanted to explore 3 different formulations using different norms.

3.1 Standard LP (L_2)

Here we are aiming to maximize the total economic benefit subject to constraints.

$$\begin{aligned} \max_x \quad & Z = 1.00x_1 + 1.60x_2 + 1.18x_3 + 1.18x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 = 1 \\ & 1.00x_1 + 0.69x_2 + 0.65x_3 + 0.65x_4 \geq 0.80 \\ & 1.00x_1 + 0.17x_2 + 0.08x_3 + 0.10x_4 \leq 0.50 \\ & x_i \geq 0 \quad \forall i \end{aligned}$$

3.2 L_1

Minimize total absolute deviation from a balanced target allocation $d = [0.25, 0.25, 0.25, 0.25]^T$.

$$\begin{aligned} \min_{x,t} \quad & \sum_{i=1}^4 t_i \\ \text{s.t.} \quad & -t_i \leq x_i - 0.25 \leq t_i, \quad i = 1, \dots, 4 \\ & x_1 + x_2 + x_3 + x_4 = 1 \\ & 1.00x_1 + 0.69x_2 + 0.65x_3 + 0.65x_4 \geq 0.80 \\ & 1.00x_1 + 0.17x_2 + 0.08x_3 + 0.10x_4 \leq 0.50 \\ & x_i \geq 0, \quad t_i \geq 0 \quad \forall i \end{aligned}$$

where $t_i = |x_i - 0.25|$ are slack variables. This gives us diversification.

3.3 L_∞

Minimize the *maximum* deviation from target.

$$\begin{aligned} \min_{x,s} \quad & s \\ \text{s.t.} \quad & -s \leq x_i - 0.25 \leq s, \quad i = 1, \dots, 4 \\ & x_1 + x_2 + x_3 + x_4 = 1 \\ & 1.00x_1 + 0.69x_2 + 0.65x_3 + 0.65x_4 \geq 0.80 \\ & 1.00x_1 + 0.17x_2 + 0.08x_3 + 0.10x_4 \leq 0.50 \\ & x_i \geq 0, \quad s \geq 0 \end{aligned}$$

where $s = \max_i |x_i - 0.25|$. This ensures maximum diversification, this simply means that no rotation gets neglected.

4 Results & Analysis

Looking at the CVXPY results (see Appendix), we get some interesting patterns. Figure 1 summarizes allocations, profits, cereal output, GHG, and deviations in one view.

Standard LP (maximize benefit) L2: This gives us $x_1 = 0.355$, $x_2 = 0.645$, $x_3 = 0$, $x_4 = 0$. Economic benefit is 1.387 which means 38.7% more than the baseline. Cereal production is exactly 0.80 (binding constraint), and GHG is 0.465 (below the 0.50 cap). The land constraint is also binding and active. Furthermore we have $x_3 = 0$ and $x_4 = 0$, so the peanut and soybean non-negativity constraints are active. Meaning that we have 4 active constraints in 3D space and so this is a vertex of the feasible polytope, which makes sense as this is exactly where we would expect the solution of a linear program to be.

The solution prioritizes SpWM (64.5%, highest profit) and includes WM (35.5%) to meet the cereal requirement. Peanut and soybean get zero allocation. This makes sense as I believe other than in cereal production they perform poorer than Sweet Potato.

L₁ solution (minimize total deviation): This gives us $x_1 = 0.400$, $x_2 = 0.250$, $x_3 = 0.176$, $x_4 = 0.174$. Economic benefit drops slightly to 1.213 (about 21.3% above baseline), but now all four rotations are being used.

The cereal constraint is still binding at 0.80. The L₁ objective trades off some SpWM allocation for more balanced distribution—notice peanut now gets 17.6% instead of zero. The L₁ norm minimizes the *sum* of all deviations from the balanced target, which means it's willing to let some crops deviate more as long as the total deviation is minimized, which is why we have peanut and soybean in the solution. What this means for us is that it diversifies risk and so we are not as dependent on any one crop performance. This has better spread than the standard LP.

L-infinity solution (minimize max deviation): This gives us $x_1 = 0.385$, $x_2 = 0.385$, $x_3 = 0.115$, $x_4 = 0.115$. Economic benefit is 1.272 (about 27.2% above baseline). Maximum deviation from any target is 0.135.

Again cereal is binding at 0.80. The L-infinity objective forces even more balance—peanut gets 11.5% and SpWM drops to 38.5%. Again this diversifies risk, in this case the L-infinity norm minimizes the *single largest* deviation, creating a more symmetrical allocation, then L₁. This has the best spread of all three solutions, it puts a ceiling on how unbalanced a single crop is and it minimize the worst deviation from targets.

Consistent patterns across all methods:

Across all 3 norm solution we have WM present in some quantity and we also have meeting cereal demand at exactly at 0.80. The differences between the models is how much SpWM versus legumes are included.

The cereal constraint is the real bottleneck. All solutions sit right on that boundary (cereal production = 0.80), which means we're trading off maximum profit for food security. The GHG constraint (0.50 cap) is slack in all 3 cases, basically we're meeting the constraint with room to spare (emissions around 0.46–0.47).

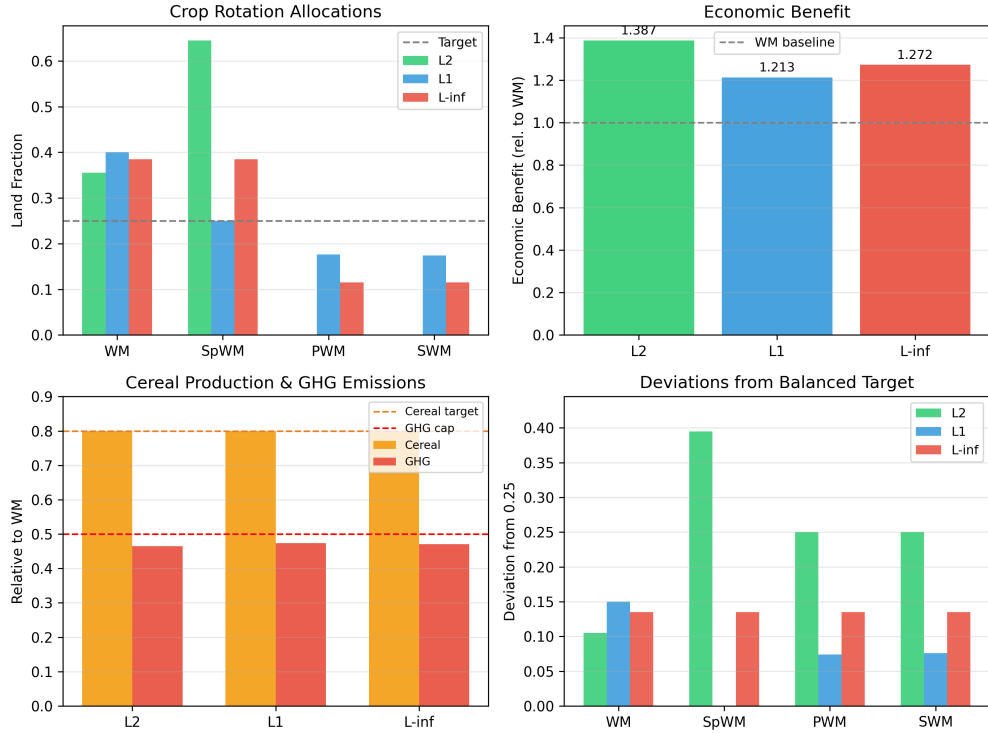


Figure 1: Four-panel summary comparing L2 (max benefit), L₁ (min total deviation), and L_∞ (min max deviation). Top-left: land allocations vs. 0.25 target line. Top-right: economic benefit vs. WM baseline. Bottom-left: cereal and GHG vs. their thresholds. Bottom-right: per-crop deviations from the balanced target.

5 Interpretation

In all 3 solutions they hit the cereal constrain at equality with 0.80, meaning that food security is a binding limit on the economic optimization. The standard LP maximizes the profit 38.7% above baseline by allocation most crops to SpWM (the most profitable crop) and the rest WM to compensate for the cereal. The other two norms sacrifice some of this profit to diversify across all four rotations, which in the real world might be interpreted as hedging against shocks like price volatility or disease outbreaks of a single crop. The L₁ approach minimizes total imbalance (good if you care about aggregate portfolio risk), and the L-infinity ensures no crop is over or under represented (I am not sure how this might be useful in the farmer context though). From a policy perspective all 3 solutions do achieve massive emission cuts (47-54%) while maintaining food security. So if you optimize correctly you can still maintain food security, GHG emissions, and profitability :)

Appendix:

AI Statement: I used GPT-5.1 to debug CVXPY syntax, generate plotting code, and help with LaTeX formatting. I also used it to correct sentence level grammar and latex formatting, but all words are my own. I did use GPT-5.1 to generate my figure caption.

References: - Yang, X., Xiong, J., Du, T., Ju, X., Gan, Y., Li, S., Xia, L., Shen, Y., Pacenka, S., Steenhuis, T. S., Siddique, K. H. M., Kang, S., & Butterbach-Bahl, K. (2024). Diversifying crop rotation increases food production, reduces net greenhouse gas emissions and improves soil health. Nature Communications, 15, Article 198. - <https://www.nature.com/articles/s41467-023-44464-9>

Code:

```
import numpy as np
import matplotlib
matplotlib.use('Agg') # headless-safe plots (no GUI needed)
import cvxpy as cp
import matplotlib.pyplot as plt

# Normalized data (relative to WM baseline)
# Rotations share the same ordering everywhere
rotations = ['WM', 'SpWM', 'PWM', 'SWM']
# Objective coefficients (economic benefit multipliers)
economic_benefit = np.array([1.00, 1.60, 1.18, 1.18])
# Constraint coefficients for cereal calories ( $\geq 0.80$ )
cereal_calories = np.array([1.00, 0.69, 0.65, 0.65])
# Constraint coefficients for GHG ( $\leq 0.50$ )
ghg_emissions = np.array([1.00, 0.17, 0.08, 0.10])
# Target allocation for balanced mix (used in L1/Linf)
target = np.array([0.25, 0.25, 0.25, 0.25])

def solve_with_fallback(obj, constraints, var):
    """Solve once with SCS, then fall back to default if needed."""
    prob = cp.Problem(obj, constraints)
    vals, status = None, None
    for solver_choice in [cp.SCS, None]:
        try:
            prob.solve(solver=solver_choice, verbose=False)
        except Exception:
            continue
        status = prob.status
        # Accept solutions that are optimal or close enough
        if var.value is not None and status in (cp.OPTIMAL, cp.OPTIMAL_INACCURATE):
            vals = np.array(var.value).flatten()
            break
    return status, vals

# ---- Standard LP (profit max) ----
x_l2 = cp.Variable(4)
```

```

status_l2, alloc_l2 = solve_with_fallback(
    cp.Maximize(economic_benefit @ x_l2),
    [
        cp.sum(x_l2) == 1,
        cereal_calories @ x_l2 >= 0.80,
        ghg_emissions @ x_l2 <= 0.50,
        x_l2 >= 0,
    ],
    x_l2,
)

# ---- L1 (minimize total deviation from target) ----
x_l1 = cp.Variable(4)
t = cp.Variable(4)
status_l1, alloc_l1 = solve_with_fallback(
    cp.Minimize(cp.sum(t)),
    [
        x_l1 - target <= t,
        x_l1 - target >= -t,
        cp.sum(x_l1) == 1,
        cereal_calories @ x_l1 >= 0.80,
        ghg_emissions @ x_l1 <= 0.50,
        x_l1 >= 0,
        t >= 0,
    ],
    x_l1,
)

# ---- L-infinity (minimize worst-case deviation) ----
x_linf = cp.Variable(4)
s = cp.Variable()
status_linf, alloc_linf = solve_with_fallback(
    cp.Minimize(s),
    [
        x_linf - target <= s,
        x_linf - target >= -s,
        cp.sum(x_linf) == 1,
        cereal_calories @ x_linf >= 0.80,
        ghg_emissions @ x_linf <= 0.50,
        x_linf >= 0,
        s >= 0,
    ],
    x_linf,
)

statuses = {"L2": status_l2, "L1": status_l1, "L-inf": status_linf}

def calc_metrics(x):
    if x is None:
        return None

```

```

    return (
        float(economic_benefit @ x),
        float(cereal_calories @ x),
        float(ghg_emissions @ x),
    )

m_l2 = calc_metrics(alloc_l2)
m_l1 = calc_metrics(alloc_l1)
m_linf = calc_metrics(alloc_linf)

if any(v is None for v in [alloc_l2, alloc_l1, alloc_linf]):
    print("One or more solves failed or are infeasible. Statuses:", statuses)
else:
    # Print results only (plotting removed per request)
    print('='*70)
    print('Statuses:', statuses)
    print('L2 (Max Benefit):')
    for i, r in enumerate(rotations):
        print(f' {r}: {alloc_l2[i]:.3f} ({alloc_l2[i]*100:.1f}%)')
    print(f' Econ: {m_l2[0]:.3f} | Cereal: {m_l2[1]:.3f} | GHG: {m_l2[2]:.3f}')

    print('\nL1 (Balanced):')
    for i, r in enumerate(rotations):
        print(f' {r}: {alloc_l1[i]:.3f} ({alloc_l1[i]*100:.1f}%)')
    print(f' Econ: {m_l1[0]:.3f} | Cereal: {m_l1[1]:.3f} | GHG: {m_l1[2]:.3f}')
    print(f' Total dev: {np.sum(dev_l1):.3f}')

    print('\nL-inf (Min-Max):')
    for i, r in enumerate(rotations):
        print(f' {r}: {alloc_linf[i]:.3f} ({alloc_linf[i]*100:.1f}%)')
    print(f' Econ: {m_linf[0]:.3f} | Cereal: {m_linf[1]:.3f} | GHG: {m_linf[2]:.3f}')
    print(f' Max dev: {np.max(dev_linf):.3f}')
    print('='*70)

```